Automata Theory

Turing Machines

CS32022

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7. Turing Machine

Non-Context-Free Languages and the Pumping Lemma

- Every context-free language possesses a pumping length p such that any string s with length at least p can be partitioned into five parts $s = uvwxy$.
- The combined middle substring vwx must have length at most p, and substrings v and x cannot both be empty ($|vx| >= 1$).
- For all non-negative integers i, the pumped string uv^iwx^iy must remain inside the context-free language.
- Assume the target language is context-free, select a representative string s, and demonstrate that every valid partition $uvwxy$ fails for some value of i.
- Languages requiring simultaneous matching across three independent blocks fail the lemma because a pushdown stack cannot track three separate counts.

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Exercise 01

Assume $L = \{ a^n b^n c^n \mid n \geq 0 \}$ is context-free. Let p be its pumping length given by the CFL Pumping Lemma.

Select $s = a^p b^p c^p \in L$. Note that $|s| = 3p \geq p$.

By Condition 2 of the Pumping Lemma, $|vwx| \leq p$. Because vwx has a maximum length of p, it cannot contain all three symbol types (a, b, and c). It can contain at most two distinct symbol types:

- **Case 1:** vwx consists entirely of a's, entirely of b's, or entirely of c's.
- **Case 2:** vwx spans the boundary between a's and b's (contains only a's and b's).
- **Case 3:** vwx spans the boundary between b's and c's (contains only b's and c's).

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Exercise 01 (Continued)

Consider $s' = uv^2wx^2y$:

- Since $|vx| \geq 1$, pumping v and x increases the count of whichever symbols appear in v and x.
- **In Case 1:** Only one symbol count increases (e.g., more a's, but b's and c's remain p).
- **In Case 2 or Case 3:** At most two symbol counts increase, leaving the third symbol count equal to p.
- **In all cases:** The numbers of a's, b's, and c's in s' are no longer equal.

Therefore, $uv^2wx^2y \notin L$, contradicting the Pumping Lemma. L is not context-free.

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Introduction to Turing Machines

- A Turing Machine uses an **infinitely long memory** tape that serves as both input medium and unlimited read/write storage.
- The read/write head can scan individual tape cells, overwrite symbols, and move freely to the left or right.
- A Turing Machine is defined as a 7-tuple containing states, input alphabet, tape alphabet, transition function, start state, accept state, and reject state.
- Entering either the accept state or the reject state causes the machine to halt its computation immediately.
- The deterministic transition function maps a current state and scanned symbol to a new state, a replacement symbol, and a head movement direction.

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Turing Machines: Formal Definition

$$\mathbf{M} = (Q, \Sigma, \Gamma, \delta, q_0, q_{\text{accept}}, q_{\text{reject}})$$

Q: Finite set of states.

Σ: Input alphabet (does not contain the blank symbol ␣).

Γ: Tape alphabet, where $\Sigma \subset \Gamma$ and $␣ \in \Gamma$.

δ: $Q \times \Gamma \rightarrow Q \times \Gamma \times \{L, R\}$: Transition function.

$q_0 \in Q$: Start state.

$q_{\text{accept}} \in Q$: Halting accept state.

$q_{\text{reject}} \in Q$: Halting reject state ($q_{\text{accept}} \neq q_{\text{reject}}$).